

Epiplexity

Definitions, estimators, follow-ups and alternatives,
with consequences for small-model estimation

Literature to 23 September 2026

Abstract

Epiplexity S_T is the length of the program that minimises a two-part code for a dataset among programs that evaluate and sample within a compute bound T (Finzi et al., 2026). The rigorous results are asymptotic and rest on cryptographic assumptions. The published values come from two neural estimators: the area of the online loss curve above the final loss; the cumulative teacher–student divergence of requential coding (Qiu et al., 2026). These notes state the definitions and theorems, compare six estimators and tabulate every trained experiment. They assess the EpiSelect and EpiGen follow-up (Su et al., 2026) and about forty papers citing the founding one, then place older measures of structure on shared criteria. For a project with multilayer perceptrons (MLPs) on small datasets and one Colab GPU, the main questions remain open. No published estimate uses an MLP. No estimate reports a spread over seeds. The held-out loss floor, the units and the per-message overhead of requential coding each shift a small-scale value by an amount comparable to the signal.

1 Setup and notation

Logarithms are base 2 and code lengths are in bits. The released code of every paper below logs natural-log losses; a value in nats becomes bits after division by $\ln 2$ (Qiu and Finzi, 2026; Qiu, 2026). $\text{KL}(p||q)$ is the Kullback–Leibler divergence in bits.

X is the dataset whose information is measured, of $\mathcal{D}$ tokens or examples; Finzi et al. (2026) call it the test dataset. A training run sees units $Z_0, \dots, Z_{M-1}$ in order; P_i is the model after i updates. When every unit is a token, $M = D$, the number of training tokens. N is the number of parameters, embeddings included in the reference code (Qiu and Finzi, 2026). Compute is measured in floating-point operations (FLOPs). Training for D tokens takes $6ND$ FLOPs and evaluating on X takes $2ND$ (Finzi et al., 2026). A run is admissible under the compute bound T when

$$6ND + 2ND \leq T. \tag{1}$$

One Colab GPU means one T4, L4 or A100 in a single session.

“(Our derivation)” marks an inference absent from the cited source; “(our measurement)” marks a value obtained by running released code on a CPU; in tables * marks either. The sign $\approx$ marks a value estimated from a published figure, with an error of roughly ten per cent.

2 Definitions and theorems

Definition 1 (T -time probabilistic model; Def. 7 of Finzi et al., 2026). Let $T : \mathbb{N} \rightarrow \mathbb{N}$ be non-decreasing and time-constructible. Let $\mathcal{U}$ be a fixed prefix-free universal Turing machine. A

prefix-free program P is a T -time probabilistic model over $\{0, 1\}^n$ if both modes of P halt within $T(n)$ steps: $\mathcal{U}(P, (0, x))$ outputs a probability $P(x)$ with a finite binary expansion; $\mathcal{U}(P, (1, u))$ outputs a sample in $\{0, 1\}^n$ from an infinite random tape u . The probabilities sum to one and the samples follow P . $\mathcal{P}_T$ is the set of such programs; $|P|$ is the length of P in bits.

$K(x)$ denotes the prefix Kolmogorov complexity, the length of the shortest program for which $\mathcal{U}$ outputs x (Li and Vitányi, 2019).

Definition 2 (epiplexity; Def. 8 of Finzi et al., 2026). For a random variable X on $\{0, 1\}^n$, let

$$P^* = \arg \min_{P \in \mathcal{P}_T} \{|P| + \mathbb{E}[\log 1/P(X)]\},$$

with ties broken by the shortest program. The epiplexity is $S_T(X) := |P^*|$. The time-bounded entropy is $H_T(X) := \mathbb{E}[\log 1/P^*(X)]$. The sum $\text{MDL}_T(X) := S_T(X) + H_T(X)$ is the total time-bounded information, a minimum description length (MDL).

Definition 3 (conditional; Def. 11 of Finzi et al., 2026). $S_T(Y | X)$ and $H_T(Y | X)$ follow from Definition 2 with $\mathbb{E}[\log 1/P(X)]$ replaced by $\mathbb{E}_{(X,Y)}[-\log P(Y | X)]$ and the minimum taken over programs whose conditional $P_{Y|x}$ belongs to $\mathcal{P}_T$ for every fixed x .

In general $H_T(Y, X) - H_T(X) \neq H_T(Y | X)$. A conditional estimate sums code length over label tokens only, while the time bound includes input tokens (Finzi et al., 2026, App. B.1). A classifier therefore estimates $S_T(Y | X)$ (our derivation). Every estimator below replaces $\mathcal{P}_T$ by the models one optimiser reaches from one architecture, a restriction the authors allow in a footnote to Def. 7.

Proposition 1 (basic properties; Sec. 3 of Finzi et al., 2026). Let c_1, c_2 be constants independent of X and $H(X)$ the Shannon entropy. (i) $S_T, H_T \geq 0$. (ii) $H(X) \leq \text{MDL}_T(X) \leq n + c_1$. (iii) $\text{MDL}_{T'} \leq \text{MDL}_T$ whenever $T' \geq T$. (iv) $\text{MDL}_{T'}(f^{-1}(X)) \leq \text{MDL}_T(X) + |f| + c_2$ with $T' = T + \text{Time}(f)$, for a program f computing a bijection in fixed time. (v) For uniform U_n and any $T(n) \geq c_1$, $S_T(U_n) \leq c_2$. The equal mixture of 0101... and 1010... has $S_T = O(1)$ and $H_T = O(1)$ under a linear time bound.

Only MDL_T is monotone in T . S_T can fall as T grows, as in the emergence experiment of Table 4; a study should report the curve $T \mapsto S_T$ (our derivation).

Table 1 collects the theorems. A pseudorandom generator (PRG) G stretches k uniform bits U_k to $n = \text{poly}(k)$ bits. No non-uniform probabilistic polynomial-time (PPT) test separates $G(U_k)$ from U_n with advantage above $\varepsilon(k)$. A one-way function (OWF) is computable in polynomial time and hard to invert for such tests. The subscript Poly means that the bound holds for every polynomial time bound $T(n)$.

Under cryptographic assumptions and polynomial time bounds the theory settles three claims. Pseudorandom data has near-maximal time-bounded entropy and near-constant epiplexity. A deterministic map can create time-bounded entropy. Time-bounded entropy depends on the order of factorisation. No theorem covers the finite, FLOP-bounded estimators of Section 3; those rest on the scaling assumptions of App. B.4 of Finzi et al. (2026).

Table 1: Theorems of Finzi et al. (2026), v2 numbering; c is a constant. Paradox 3 has a definition and a conjecture but no theorem.

Result	Assumption	Statement
Thm 9 (PRG)	PRG G with advantage $\varepsilon(k)$	$n - 2 - n\varepsilon(k) < H_{\text{Poly}}(G(U_k)) \leq n + c$; $S_{\text{Poly}}(G(U_k)) \leq c + n\varepsilon(k)$; Shannon entropy k
Thm 10	OWFs secure against non-uniform PPT	some X_n has $S_{\text{Poly}}(X_n) = \Omega(\log n)$; non-constructive
Thm 12 (Paradox 1)	as Thm 9	$H_{\text{Poly}}(G(U_k)) - H_{\text{Poly}}(U_k) > n - k - n\varepsilon(k) - c$
Thm 13 (Paradox 2)	one-way permutation f ; $X = U_n$; $Y = f(X)$	$H_{\text{Poly}}(X Y) + H_{\text{Poly}}(Y) > H_{\text{Poly}}(Y X) + H_{\text{Poly}}(X) + \omega(\log n)$; by Cor. 26 no polynomial-time model fitting the forward direction obeys Bayes' rule
Def. 14 (Paradox 3)	none	emergence: $S_{T_1} - S_{T_2} = \Theta(1)$ for one step, $\omega(1)$ for $k(n)$ steps, $T_1 = o(T_2)$; no instance proven
Thm 30; App. B.4	loss C^2 with a unique optimum, larger models more sample-efficient	optimal $N^*(T)$ and $D^*(T)$ increase with T ; $D^* \rightarrow \mathcal{D}$ from below and $N^* \sim T/(8\mathcal{D})$; with infinite compute S_∞ is non-decreasing in $\mathcal{D}$

Does any explicit distribution have epiplexity above $\Omega(\log n)$? Does computation create epiplexity, as opposed to entropy? Reason: the existence proof is non-constructive; Theorem 12 concerns H_T ; the S_T claims of Paradoxes 1 and 3 rest on the cellular-automaton and looped-model experiments; no instance of Definition 14 is proven (Finzi et al., 2026).

3 Estimators

3.1 Prequential

Prequential coding (Dawid, 1984) sends Z_i with $\log 1/P_i(Z_i)$ bits and then trains on it. The total encodes the data together with the final model and decodes in $6ND$ FLOPs. Following Zhang et al. (2020) and Finzi et al. (2025), Eq. 8 of Finzi et al. (2026) estimates the model part as

$$|\text{P}_{\text{preq}}| \approx \sum_{i=0}^{M-1} [\log 1/P_i(Z_i) - \log 1/P_M(Z_i)]. \quad (2)$$

The right side is the area of the online loss curve above the final loss. The two-part code adds $\mathbb{E}[\log 1/P_M(X)]$, estimated as the held-out loss per token times $\mathcal{D}$. The authors give two reasons why (2) is not rigorous. The difference of two upper bounds on Kolmogorov complexities does not bound $K(P_M)$. Symmetry of information fails for time-bounded complexity, which leaves the decoding time unproven.

Every experiment except the comparison with adaptive data optimisation (ADO; Jiang et al., 2025) replaces $\sum_i \log 1/P_M(Z_i)$ by M times the loss of P_M on unseen data (Finzi et al., 2026, App. B.1). The replacement assumes one pass over i.i.d. data with a small generalisation gap. Zhang et al. (2020) compute the same difference on examples outside the training set (information transfer); on CIFAR-10 with random labels it remains near zero although training accuracy is high. Let p_{data} be the data distribution. By Eq. 1 of Qiu et al. (2026), the expectation of (2) with a held-out floor is

(our derivation)

$$\mathbb{E}|\mathbf{P}_{\text{req}}| = \sum_{i=0}^{M-1} [\text{KL}(p_{\text{data}} \| P_i) - \text{KL}(p_{\text{data}} \| P_M)]. \quad (3)$$

The data entropy cancels. The raw prequential code $\sum_i \log 1/P_i(Z_i)$ keeps it and grows linearly after learning stops (Qiu et al., 2026).

3.2 Requential

Requential coding transmits a student model (Qiu et al., 2026). Encoder and decoder share the student initialisation, update rule, pseudorandom seed, number of steps and batch size. At step i both sides generate proposals from the student P_i^s . The encoder holds a teacher P_i^t trained on real data. By relative entropy coding (REC) it selects a proposal distributed as P_i^t and sends the index with an Elias-delta code. Both sides then train the student on that batch. Only hard samples are coded. With $\text{KL}_i = \text{KL}(P_i^t \| P_i^s)$ and the sum over student steps, Eq. 2 of Qiu et al. (2026) bounds the expected length:

$$|\mathbf{P}_{\text{req}}| \leq \sum_i [\text{KL}_i + 2 \log(1 + \text{KL}_i) + \kappa], \quad \kappa < 5.21. \quad (4)$$

Eq. 9 of Finzi et al. (2026) has $\log(1 + \text{KL}_i) + 4$ in place of the last two terms; the factor 2 comes from coding the index with a universal integer code. Both papers report $\sum_i \text{KL}_i$. The remaining terms add 5 to 15 bits per step and are negligible only when every KL_i is far above that (our derivation).

Measuring the length needs no REC search: sampling from the teacher gives the divergences at about 7/3 of the FLOPs of training the teacher alone. Transmission would need about 2^{KL_i} proposals per message; no encoder is released (Qiu et al., 2026).

The two papers keep KL_i small in different ways. The founding paper distils from an exponential moving average (EMA) of the teacher and freezes the teacher while KL_i exceeds a threshold. The later paper keeps the EMA and adds iso-loss projection: at steps $100 \cdot 1.5^j$ the teacher is reset to the student and retrained on real data, with the student paused, until it recovers the earlier validation loss. With a static teacher close to p_{data} , requential coding reduces to (2), which the founding authors therefore expect to exceed the requential code (Finzi et al., 2026, App. B.2).

For a classifier over C classes with batch inputs x_b known to the decoder, $\text{KL}_i = \sum_b \sum_{c=1}^C P_i^t(c | x_b) \log[P_i^t(c | x_b)/P_i^s(c | x_b)]$ is exact and needs one forward pass of each network. Sampling a label replaces autoregressive generation (our derivation).

3.3 From a code length to S_T

Finzi et al. (2026, App. B.1) obtain S_T from a sweep. They tune the learning rate on a small model and transfer it by μP and CompleteP. They train a grid of widths and depths, each for more tokens than $\mathcal{D}$, with a constant learning rate and an EMA of the iterates; every logged checkpoint becomes a point (compute of (1), $|\mathbf{P}| + \mathbb{E} \log 1/P(X)$). S_T is the model part at compute T on the lower convex hull of all points, of which only the median point per run is kept. The grid has 30 configurations per cellular-automaton rule and 48 per natural dataset, each with one seed (Qiu and Finzi, 2026). When a model reaches a known loss floor, as in the induction tasks, one model per condition suffices. For ranking datasets a single run of (2) is the crude estimate the authors recommend. They expect the hull heuristic, the fixed architecture and imperfect hyperparameters to cause only sub-leading errors.

Proposition 2 (power-law loss; App. B.3 of Finzi et al., 2026). Let the loss after D tokens with N parameters be $\mathcal{L}(N, D) = E + (N_0/N)^\alpha + (D_0/D)^\beta$ with $0 < \alpha, \beta < 1$. Write $\hat{N} = N/N_0$, $\hat{D} = D/D_0$, $\hat{\mathcal{D}} = \mathcal{D}/D_0$ and $\hat{T} = T/(2N_0D_0)$; equality in (1) becomes $\hat{T} = \hat{N}(3\hat{D} + \hat{\mathcal{D}})$. Take (2) as model part and $\mathcal{DL}$ as data part. Then:

- (a) $|\mathbf{P}_{\text{preq}}| = \frac{\beta}{1-\beta} D_0 \hat{D}^{1-\beta}$ for large D ;
- (b) the optimum solves $\beta \hat{D}^{-\beta-1}(\hat{\mathcal{D}} - \hat{D}) = 3\alpha \hat{\mathcal{D}} \hat{T}^{-\alpha}(3\hat{D} + \hat{\mathcal{D}})^{\alpha-1}$ with $\hat{N}^* = \hat{T}/(3\hat{D}^* + \hat{\mathcal{D}})$;
- (c) as $T \rightarrow \infty$, $\hat{D}^* \rightarrow \hat{\mathcal{D}}$ and $\hat{N}^* \sim \hat{T}/(4\hat{\mathcal{D}})$, with limits $S_\infty(X) = \frac{\beta}{1-\beta} D_0^\beta \mathcal{D}^{1-\beta}$ and $H_\infty(X) = \mathcal{DE} + D_0^\beta \mathcal{D}^{1-\beta}$;
- (d) for $\hat{D}^* \ll \hat{\mathcal{D}}$, $\hat{N}^* \approx \hat{T}/\hat{\mathcal{D}}$, $S_T \propto T^{\alpha(1-\beta)/(1+\beta)}$ and $H_T - \mathcal{DE} \propto T^{-\alpha\beta/(1+\beta)}$.

Code lengths are in the units of E , nats in the fitted laws. With the Chinchilla exponents $\alpha \approx 0.34$, $\beta \approx 0.28$ the exponents in (d) are 0.19 and -0.07 . For language ($\beta = 0.285$, $D_0 = 1.49 \times 10^9$, $\mathcal{D} = 10^{12}$), part (c) gives 9.0×10^{10} bits, the value of Fig. 8b there (our derivation).

3.4 Alternative estimators from 2026

Excess description length (EDL). Donoway et al. (2026) posted EDL two days after epileptixity; a single-author version appeared at ISIT 2026 (Donoway, 2026). Let $\ell(\theta; x, y) = -\log p_\theta(y | x)$, $L(\theta) = \mathbb{E}_{p_{\text{data}}} \ell$ the population loss, $\mathcal{A}$ the training algorithm, θ_0 the initial parameters and θ_T the parameters after compute T (steps, epochs or FLOPs). For a sample $S = (z_1, \dots, z_M)$ the ISIT definition is

$$\text{EDL}_T(\mathcal{A}, M) = \mathbb{E}_{S \sim p_{\text{data}}^M} \left[\mathbb{E}_\pi \sum_{i=1}^M \ell(\theta_{i-1}^\pi; z_{\pi(i)}) - M L(\theta_T) \right]. \quad (5)$$

Here π is a uniform permutation and θ_{i-1}^π are the parameters after $i-1$ examples of the first pass; later epochs enter only through $L(\theta_T)$. The ISIT version proves non-negativity, monotonicity in compute, additivity, saturation at M times the input–label mutual information $I(X; Y)$ and $\text{EDL}_T \lesssim 0$ for random labels. Neither version reports a neural-network experiment or code. EDL evaluates a given algorithm, subtracts a population loss and allows compute beyond one pass (Donoway, 2026). For one pass over i.i.d. data it equals the expectation of (2) with the population loss as floor (our derivation). The supremum over algorithms is not a structure measure: an algorithm idle during the first pass reaches $M[L(\theta_0) - L(\theta_T)]$, linear in M even for a one-line rule (our derivation).

Reservoir score. Zhang and Levin (2026) replace training by a closed form. A frozen random network ϕ maps N_s inputs to m standardised features $\tilde{H} \in \mathbb{R}^{N_s \times m}$; $\tilde{Y}$ are the targets, centred and divided by a fixed scale. A ridge-regression head with penalty λ , $W_\lambda = (\tilde{H}^\top \tilde{H} + \lambda I_m)^{-1} \tilde{H}^\top \tilde{Y}$, with singular values s_i and a resolution η (1, or 30 for MNIST) gives

$$S^\phi(Y | X) = \frac{1}{2} \sum_i \log_2(1 + \eta s_i^2). \quad (6)$$

Ridge approximates the two-part minimiser; the exact minimiser ranks automaton rules with Spearman correlation 0.997. The score is capacity-bounded, not time-bounded, with the prefactor fixed at $\frac{1}{2}$ by convention; the bits of S^ϕ are not commensurable with S_T (our derivation). On elementary cellular automata (ECA) it ranks rule 110 first and rule 54 above rule 30, an order that reverses once the receptive field exceeds radius two. The authors claim that noise contributes nothing to the model part. Table 3 shows a positive floor at finite N_s instead: independent targets

score close to rule 15 and decay slowly with N_s (our measurement). Rule 30 scores above rule 15, opposite to Finzi et al. (2026); both exceed the noise floor by a few bits only. Comparisons need a shuffled-target baseline at matched N_s . Three independent repositories reused the score within two months (lcrh, 2026; the-puzzler, 2026; eberlful, 2026).

Small-data prequential estimate. Liu et al. (2026) adapt (2) to multi-epoch fine-tuning on small sets (Liu, 2026). The online loss L_{online} over the N_{train} tokens of the first epoch is frozen. After each epoch the model is evaluated on all training examples (L_{train}) and on a validation split (L_{val} over N_{val} tokens). The estimate is $S = (L_{\text{online}} - L_{\text{train}}) / \ln 2$ at the epoch minimising $S/N_{\text{train}} + L_{\text{val}}/(N_{\text{val}} \ln 2)$. With $M = N_{\text{train}}$ units, L_{train} is in-sample, which gives $S_{\text{Liu}} = \text{EDL}_T + M [L(\theta_T) - L_{\text{train}}/M]$ in expectation (our derivation). Memorisation raises it; only the epoch choice limits it. The code evaluates online losses with dropout active and final losses without it, which biases S upward (our derivation from the code).

Finite-time thermodynamics. Ohzeki (2026) extends the widely applicable Bayesian information criterion to finite-time stochastic-gradient Langevin ensembles. The result is a computable thermodynamic term of time-bounded MDL, which the author states is not epiplexity itself. The paper is theoretical.

Table 2 compares the six estimators on the properties that define epiplexity.

Table 2: Estimators of learnable structure. Entries without * are claims of the cited papers.

Estimator	Zero on noise	Zero on trivial data	Compute-bounded	Minimises over models	Training runs	Code
Prequential (2); held-out variant (Zhang et al., 2020)	with a held-out or one-pass floor*	yes	yes	sweep, hull	5–30 per grid of T ; 1 to rank	PyTorch, JAX (Qiu and Finzi, 2026)
Requential (4)	yes, Table 5	yes, Table 5	yes	founding paper only	$\approx 7/3$ of a run per configuration; 2–10 times prequential	JAX (Qiu, 2026)
EDL (5)	$\lesssim 0$, proved	zero with calibrated start	indexed by T	no	1 plus held-out loss	none
Reservoir S^ϕ (6)	no at finite N_s^*	zero on constants*	no	no	none; < 1 s on a CPU*	PyTorch (Zhang, 2026)
S_{Liu} (Liu et al., 2026)	via epoch choice*	yes*	no	epochs only	1 multi-epoch run	PyTorch (Liu, 2026)
Free energy (Ohzeki, 2026)	not stated	not stated	finite time	not stated	Langevin ensemble	none

4 Experiments of the founding and requential papers

Table 4 lists every trained experiment of Finzi et al. (2026) and Qiu et al. (2026). All use GPT-2-style transformers trained with Adam and μP learning-rate transfer; neither paper trains an MLP or a convolutional network. The founding paper used six 2080 Ti GPUs for small experiments, six Titan RTX GPUs and 32 TPU v4 chips for natural data. The requential paper used TPU v6e-8 hosts.

Table 3: Reservoir score S^ϕ in bits (our measurement with the released code of Zhang, 2026, CPU). Configuration: depth 3, 256 channels, kernel 3, $\lambda = 0.03$, $\eta = 1$, target window 32. Noise: targets independent of inputs. One reservoir seed per row except the last, which gives mean $\pm$ s.d. over three seeds.

N_s	rule 110	rule 54	rule 30	rule 15	noise
128	62.4	38.7	34.4	29.9	28.4
512	63.2	36.6	24.9	20.3	17.6
2048	63.7	35.5	16.8	11.0	9.5
512, three seeds	63.9 ± 0.6	37.1 ± 0.4	25.5 ± 1.0	20.9 ± 0.6	18.0 ± 1.3

Neither reports GPU-hours. Every sweep except the one-way-function experiment (two seeds) uses one seed; no figure has error bars (Qiu and Finzi, 2026; Qiu, 2026).

The ECA frontier is the smallest demonstration of the ordering structured above trivial above pseudorandom. Order dependence of S_T is measured only on chess; the small rule-30 experiment reports H_T only. In Table 5 the raw prequential code ranks uniform noise highest and the requential code ranks it lowest. The ratio of heuristic to requential code changes several-fold with the teacher recipe, which makes it a property of the recipe as much as of the data (our derivation).

5 EpiSelect and EpiGen

Su et al. (2026) present both methods in one paper by the same group (arXiv v1, 12 August 2026). Both replace the final-loss floor of (2) by the current loss, which makes the quantity optimisable online. Neither searches over (N, D) at fixed T ; the authors call the result “a tractable proxy” and leave the link to S_T open.

EpiSelect reweights data domains during pretraining. Let $L_j(s)$ be the training loss of domain j at step s and D_k the number of tokens seen from domain k . The online proxy, a cross-domain scaling law, the gradient and the sampling rule are Eqs. 2–5 there:

$$\begin{aligned} \hat{S}(t) &= \sum_j \sum_{s=1}^t [L_j(s) - L_j(t)], & \hat{L}_j &= \epsilon_j + b_j \left(\sum_k \gamma_{jk} D_k \right)^{-a_j}, \\ \frac{\partial \hat{S}}{\partial D_k} &= \sum_j \frac{a_j \gamma_{jk} D_j}{\sum_i \gamma_{ji} D_i} (\hat{L}_j - \epsilon_j), & \pi_k &\propto \exp\left(\frac{1}{\tau} \frac{\partial \hat{S}}{\partial D_k}\right). \end{aligned} \quad (7)$$

The weights $\gamma_{jk} > 0$ with $\sum_k \gamma_{jk} = 1$ couple domains, ϵ_j is the irreducible loss, π_k is the sampling probability of domain k and $\tau = 1$. EpiSelect refits the law every 1000 steps from 729 L-BFGS initialisations and averages weights with momentum 0.9 (Su, 2026b). With γ the identity the gradient reduces to $a_k(\hat{L}_k - \epsilon_k)$: the ADO slope (Jiang et al., 2025) multiplied by D_k , without prior and credit terms (our derivation).

EpiGen fine-tunes a GPT-2 generator by REINFORCE. The reward for a generated batch is the drop in learner loss on a buffer $\mathcal{B}$ of past batches after 10 learner steps on the new batch. With θ_t the learner parameters after update t , Eq. 6 there is

$$r_t = \sum_{x \in \mathcal{B}} [\log 1/P_{\theta_{t-1}}(x) - \log 1/P_{\theta_t}(x)]. \quad (8)$$

Table 4: Trained experiments. $\approx$: estimated from figures. FLOPs with *: $6ND$ computed by us. Last column: our estimate at 5–20 TFLOP/s sustained, not a measurement.

Experiment	Models	Data	FLOPs	Main result	Colab
<i>Finzi et al. (2026)</i>					
ECA rules 15, 30, 54; $Y = F^{48}(X)$, 64 cells (Fig. 3)	width 16–512, depth 1–9	$\mathcal{D} = 10^8$ label tokens	$\approx 10^{11}$ – 2×10^{17}	S_T : rule 54 $\approx 5.4 \times 10^6$ bits, rule 15 $\approx 2 \times 10^5$, rule 30 ≈ 0 ; order settles from $\approx 10^{13}$ FLOPs	width ≤ 128
Ten ECA rules (Fig. 2c)	width 16–128, depth 1–3	$\mathcal{D} = 2.5 \times 10^8$	$\leq 10^4$ steps	prequential several times requential; correlated within groups	yes
Rule 30 as one-way function (Fig. 4a)	8 layers, width 128–192	$n = 16$ –64 cells	$\approx 10^{16}$ – 10^{17}	H_T only: reverse gap grows with n	$n \leq 32$
Chess move order; transfer (Figs. 4c, 7)	10^6 – 1.6×10^8	5×10^9 chars	$\leq 5 \times 10^{18}$	reverse order: higher H_T and S_T (≈ 2.8 against 2.4×10^8 bits); centipawn accuracy ≈ 0.28 against 0.21	no
Hard induction, h hidden bits (Fig. 5b)	3 layers, width 256	$\approx 2 \times 10^9$ tokens	$\leq 2 \times 10^{16}$	S rises with h , ≈ 2.5 to 5.5×10^6 (nats)	1–3 hours each minutes
Easy induction, Markov chains (Fig. 5c)	3 layers, width 128	$\approx 6 \times 10^8$ tokens	$\leq 2.5 \times 10^{15}$	S peaks at intermediate h : $\approx 10^6$ at $h = 0$, 4.7×10^6 at $h = 6$	
Emergence, rule 54, looped (Fig. 6)	width 16–128, depth 1–32, 1–16 loops	10^9 tokens per run	$\leq 4 \times 10^{16}$ per run*	S_T and MDL_T drop abruptly when the looped model wins; peak $\approx 1.45 \times 10^7$ bits	subset
Natural data (Fig. 8a)	$\leq 1.6 \times 10^8$	5×10^9 tokens	$\leq 6 \times 10^{18}$	S_T : text $\approx 3 \times 10^8$, chess ≈ 2 – 2.8×10^8 , CIFAR-5M $\approx 9 \times 10^7$ bits	no
Scaling laws (Fig. 8b)	published fits	$\mathcal{D} = 10^{12}$	$T = 10^{25}$	language above images above video	CPU
ADO sampling (Fig. 8c)	1.3×10^9	1.25×10^{11}	$\approx 10^{21}$ *	ADO ≈ 2.45 against 1.95×10^{10} bits	no
<i>Qiu et al. (2026)</i>					
Codes compared; learnable information (Figs. 4, 9)	$\approx 10^8$	2–20 $\times 10^9$	$\approx 10^{18}$ – 10^{19} *	Table 5	toy runs
Scaling, bounds, many epochs (Figs. 5, 7, 8)	1.7×10^6 – 10^9	10^8 – 2×10^{10}	$\leq 10^{20}$ *	code for a fixed loss falls with N ; bound below 4-bit quantisation; best bound near one epoch	no

Table 5: Code lengths in bits digitised from Figs. 4 and 9 of Qiu et al. (2026); about 10^8 parameters. Heuristic: (2). Best: EMA smoothing and iso-loss projection.

Fig.	Data	Prequential, raw	Heuristic	Requential, plain	Requential, best
4	OpenWebText (characters)	3.8×10^9	9.0×10^8	1.0×10^9	7.0×10^7
4	CIFAR-5M (pixels)	9.8×10^9	2.3×10^8	4.9×10^8	3.3×10^7
4	FineWeb (BPE)	1.3×10^{10}	1.75×10^9	1.7×10^9	3.2×10^8
9	uniform random tokens	3.27×10^{10}	–	–	1.4×10^5
9	one repeated token	1.7×10^8	–	–	1.0×10^6
9	CIFAR-5M	2.3×10^{10}	–	–	3.6×10^7
9	FineWeb	2.8×10^{10}	–	–	3.6×10^8

By design, memorising one noise batch does not lower loss on the rest of the buffer. Table 6 gives the results.

Table 6: EpiSelect: mean zero-shot accuracy over 10 LM-Eval tasks after 15.7B Common Pile tokens. EpiGen: mean over 7 GLUE tasks and OpenWebText perplexity. From Tables 1–3 of Su et al. (2026); one seed per entry. Columns with * drop the two tasks with the largest gains (our computation).

EpiSelect	124M	1.3B	124M without BBQ, BoolQ*	1.3B without BBQ, BoolQ*
Natural sampling	.377	.422	.362	.413
ADO	.379	.425	.367	.417
EpiSelect	.394	.431	.368	.420

EpiGen (GPT-2 small)	GLUE	GLUE without CoLA, RTE*	perplexity
Pretrained	.743	.860	24.64
Frozen generator	.759	–	26.49
Perplexity reward	.756	–	104.20
No buffer	.757	–	33.10
EpiGen	.770	.858	27.58

The evidence has four limits.

1. Every entry of Table 6 comes from one seed (Su, 2026b,a). The gains concentrate in two tasks per method; without them EpiSelect and ADO differ by about a tenth of a point at 124M, while EpiGen no longer exceeds the pretrained model (our computation).
2. The motivating correlation between prequential epiplexity and mean out-of-distribution (OOD) accuracy (Pearson 0.88, $p \approx 0.05$) rests on five Pile domains. The authors kept only large domains because multi-epoch training violates the assumption of (2).
3. From a random initialisation EpiGen changes the GLUE mean by -0.003 . No learner improves on pretrained perplexity.
4. Paper and code disagree. The paper uses 30 domains and 29.7B tokens, the repository 31 and 43.3B. The EpiGen reward uses a copy of the learner with a fresh AdamW state; each GPU trains a separate learner; losses are averaged per token instead of summed as in (8). The paper states four L40S GPUs for 10 hours per run, the README two GPUs for 24 hours (Su, 2026b,a).

6 Citing work and critiques

Semantic Scholar lists 37 papers citing Finzi et al. (2026) by 23 September 2026 (Semantic Scholar, 2026); at least two more exist (Ohzeki, 2026; Okanohara, 2026). The founding paper has no venue; author pages list it as an arXiv preprint. The idea first appeared in Jiang (2025), which reports the ECA ranking stable across Mamba, temporal-convolution and transformer observers. Table 7 groups the substantive users.

Objections without a resolution.

- *Task relevance.* S_T includes structure irrelevant to any downstream task. Li (2026) constructs datasets whose total-structure proxy ranks opposite to OOD accuracy; an unreviewed reinforcement-learning curriculum weighted by epiplexity finds no correlation with transfer (sunnydigital, 2026). Reason: no task-conditioned variant has been tested.

Table 7: Papers that use epiplexity, grouped by relevance to MLP estimation on small datasets. The remaining citing papers mention it in one line (Semantic Scholar, 2026).

Relevance	Work	Use of epiplexity
High	Zhang and Levin (2026)	closed-form score (6); ECA ranking; objective for automata and an MNIST encoder; intrinsic reward in reinforcement learning, better than task reward on 9 of 10 tasks
	Qiu et al. (2026)	rigorous estimator (4)
	Donoway et al. (2026); Donoway (2026)	concurrent EDL (5), compared with S_T
	Li (2026)	proposition and synthetic experiment: a total-structure proxy ranks two pretraining datasets opposite to OOD accuracy (2 of 3 seeds)
Medium	Liu et al. (2026)	small-data prequential estimate with epoch selection
	Dineen et al. (2026)	low-rank adapter as observer, bits per token, epoch chosen by MDL
	Majchrowska and Teare (2026)	ranks self-supervised runs; estimator not described
	Ohzeki (2026)	finite-time information criterion
	Lee et al. (2026); Ren (2026)	motivation for automaton pretraining; learner improvement as generation objective
	Takahashi and Hayashi (2026)	epiplexity per joule; values benchmark-relative
Low	Okanohara (2026); Noguer i Alonso (2026); Moriondo and Azizi (2026); Hu (2026); Kim et al. (2026); Panaganti et al. (2026); Cornelisse et al. (2026); Herrmann and Schmidhuber (2026)	interpretive or motivating use; no computation

- *Looseness of (2)*. The founding group calls it a non-rigorous heuristic (Qiu et al., 2026). Reason: the ratio to the requential code depends on the teacher recipe.
- *Finite data*. The one-pass assumption fails on small datasets (Liu et al., 2026). Reason: the published fixes of Section 3.4 differ by the generalisation gap.
- *Practicality and relativity*. Each value needs a training sweep, has no gradient with respect to the data and is benchmark-relative (Zhang and Levin, 2026; Cornelisse et al., 2026; Takahashi and Hayashi, 2026). Reason: only the reservoir score is differentiable, with a different observer.

7 Theoretical alternatives

Table 8 compares older measures on the criteria that define epiplexity; Table 9 covers code lengths computed from training runs. Measures that stay near zero on random data measure structure; the others measure randomness.

Relation of each family to S_T . The structure function and the minimal sufficient statistic split an individual string into model bits and noise bits; S_T is their time-bounded, distributional analogue (our derivation). Effective complexity is closest: the total information $K(E) + H(E)$ of an ensemble E is the unbounded form of $|P| + \mathbb{E} \log 1/P(X)$ (Ay et al., 2010). Sophistication and facticity are uncomputable; naive time-bounded sophistication collapses to a constant (Lemma 29 of Finzi et al., 2026). Depth measures production time instead of model size. Excess entropy is the area between finite-block entropy rates and the asymptotic rate, the unbounded analogue of (2) (Finzi et al.,

Table 8: Measures of information and structure. *: our inference from the definition, not stated by the source.

Measure	Bounded observer	≈ 0 on random	≈ 0 on constant	Estimable	Reference
Epiplexity S_T	time, training and evaluation	yes	yes	yes	Finzi et al. (2026)
Time-bounded entropy H_T	same	no	yes	held-out loss	Finzi et al. (2026)
Kolmogorov K ; time-bounded K^t , Levin Kt	no; time for K^t , Kt	no	yes*	compressor proxy; exponential time	Li and Vitányi (2019); Levin (1973); Allender et al. (2011)
Structure function, minimal sufficient statistic; effective complexity	no	yes	yes*	upper semicomputable at best	Vereshchagin and Vitányi (2004); Gács et al. (2001); Gell-Mann and Lloyd (1996); Ay et al. (2010)
Sophistication (Koppel, naive, coarse)	no; naive time bound collapses	yes	yes*	no	Koppel (1987); Antunes and Fortnow (2009); Mota et al. (2013)
Logical depth; computational depth	no; they measure time	yes	yes*	no	Bennett (1988); Antunes et al. (2006)
Facticity	no	yes	small	no	Adriaans (2009, 2012)
Statistical complexity; excess entropy; predictive information	no	yes	yes	block entropies, discrete processes	Shalizi and Crutchfield (2001); Crutchfield and Feldman (2003); Bialek et al. (2001)
HILL, metric, Yao, next-block pseudoentropy	class of tests	no	yes*	no	Håstad et al. (1999); Barak et al. (2003); Haitner et al. (2013); Vadhan and Zheng (2012)
V-entropy	family $\mathcal{V}$, inference only	no	yes	yes	Xu et al. (2020)
V-information; pointwise V-information	family $\mathcal{V}$	if $X \perp Y$	yes*	yes	Xu et al. (2020); Ethayarajh et al. (2022)
Two-part MDL, model part	model class	yes*	yes*	simple classes	Rissanen (1978); Grünwald (2007)
Information in the weights	model class	at $\beta = 1^*$	yes*	Gaussian or Fisher	Achille et al. (2019, 2021)

2026); for such unbounded observers a pseudorandom stream has large statistical complexity (our derivation). Pseudoentropies are analogues of H_T ; next-block pseudoentropy depends on block order, a precedent for Paradox 2 (our derivation). V-entropy with $\mathcal{V} = \mathcal{P}_T$ is H_T without the model term (our derivation). V-information measures how much X helps predict Y : a simple, highly predictive rule has high V-information and low S_T . Information in the weights (Achille et al., 2021) is the closest machine-learning precursor: a divergence to a prior replaces $|\mathcal{P}|$; no time bound enters.

Remark 1 (bibliographic corrections). The founding bibliography lists “Sophistication revisited” with four authors; it is by Antunes and Fortnow alone (Antunes and Fortnow, 2009), while the four-author paper is “Computational depth” (Antunes et al., 2006). Coarse sophistication equals busy-beaver depth up to $O(\log n)$ by Theorem 5.3 of Antunes and Fortnow (2009); Ay et al. (2010)

relate effective complexity, not sophistication, to logical depth. Definition 5 of the founding paper omits $x \in S$ (issue #1 of Qiu and Finzi, 2026).

8 Practical neural-network scores

Table 9 sorts practical scores by behaviour on random and trivial labels. Four behave like S_T : the area above a held-out floor; the sequential code; surplus description length (SDL) with the floor at the irreducible loss; per-example learnability. Raw code lengths and difficulty scores rise with label noise. Consistency and confidence scores are high on trivial data.

Table 9: Practical scores by behaviour on random and trivial labels (our classification of behaviours reported in the cited papers). C : number of classes.

Score	Class	Random labels	Trivial labels	Runs	Reference
Area above held-out floor; (2) in one pass	structure	≈ 0	≈ 0	1	Zhang et al. (2020)
Sequential code	structure	≈ 0	≈ 0	teacher, student	Qiu et al. (2026)
SDL, floor at irreducible loss	structure	≈ 0	≈ 0	loss-data curve	Whitney et al. (2020)
RHO-LOSS, JEST learnability, per example	structure not yet learnt	≈ 0 or negative	≈ 0 once learnt	1 plus reference	Mindermann et al. (2022); Evans et al. (2024)
Total prequential code; gzip; bits per character	total information	$\approx M \log C$; high	≈ 0	1 or none	Blier and Ollivier (2018); Voita and Titov (2020); Pandey (2024); Delétang et al. (2024)
Area above training floor after many epochs; intrinsic dimension on training accuracy	capacity incl. memorisation	high	low	1 or many	Zhang et al. (2020); Li et al. (2018)
EL2N, GraNd, forgetting, gradient variance, margin area	difficulty	high	low	1–10	Paul et al. (2021); Toneva et al. (2019); Agarwal et al. (2022); Pleiss et al. (2020)
C-score; cartography confidence; pointwise V-information	ease	low	high	1 to thousands	Jiang et al. (2021); Swayamdipta et al. (2020); Ethayarajh et al. (2022)
V-information	predictable information	≈ 0	≈ 0	2	Xu et al. (2020)

(2) has antecedents the founding paper does not cite. Blier and Ollivier (2018) interpret the prequential code minus the log-loss of the final model as “the amount of information that the trained parameters contain about the data”; Voita and Titov (2020) use the same difference. A text search of v2 finds neither, nor Bornschein et al. (2022), Perez et al. (2021), Yogatama et al. (2019), RHO-LOSS, pointwise V-information, EL2N, Achille et al. (2021) or Bialek et al. (2001). The paper adopts the held-out heuristic of Zhang et al. (2020), “more analogous to the spirit of epiplexity”; it also cites SDL. Variance reduction transfers directly: incremental training with replay and temperature calibration (Bornschein et al., 2022); switching between models against early-block

overfitting (Blier and Ollivier, 2018); several seeds (Perez et al., 2021).

9 Consequences for small models on one GPU

No published work estimates epiplexity with an MLP or a convolutional network; every trained experiment of the founding group uses transformers (Finzi et al., 2026; Qiu et al., 2026; Su et al., 2026). Several ingredients already exist at small scale (Table 10).

Table 10: The smallest published or measured settings relevant to MLP estimation.

Setting	Size	Compute or time	Value
ECA frontier, smallest models (Finzi et al., 2026)	$\approx 3 \times 10^3$ parameters	order settles from $\approx 10^{13}$ FLOPs	$S_T \approx 10^5\text{--}10^7$ bits
Prequential code of MNIST, CIFAR-10 labels (Blier and Ollivier, 2018)	6×10^4 , 5×10^4 labels	not reported	4.10, 45.3 kbits; uniform code 199, 166
MNIST LeNet with replay (Bornschein et al., 2022)	6×10^4 labels	1.4×10^{13} FLOPs	4.4×10^3 nats
MLP classifiers on frozen MNIST features (Whitney et al., 2020)	20 sizes, 8 seeds	≈ 2 min, one GPU	loss-data curves
Reservoir score, one ECA rule*	$N_s = 512$	≈ 0.7 s, CPU	Table 3
Released PyTorch trainer, rule 54, 1 layer, width 32*	200 steps	143 s, CPU	area $\approx 8.3 \times 10^5$ nats

Pitfalls. Seven problems decide whether a small-scale value means anything.

1. *Training-set floor.* After several epochs a training-set floor turns memorised noise into apparent structure. Random labels on the $M = 6 \times 10^4$ MNIST images would then score up to $M \log_2 10 \approx 199$ kbits instead of zero (our derivation). A held-out floor keeps (2) near zero (Zhang et al., 2020); EDL makes this a theorem.
2. *Floor error times M .* A bias δ in the final loss shifts (2) by $M\delta$. At the ECA scale, $\delta = 10^{-3}$ nats over 2.5×10^8 tokens gives 3.6×10^5 bits, more than S_T of rule 15 (our derivation).
3. *Magnitudes.* The MNIST prequential code bounds the area above any non-negative floor (Table 10); Proposition 2(c) scales S_∞ as $\mathcal{D}^{1-\beta}$. Small datasets give kilobits, against megabits for the synthetic tasks of Table 4 (our derivation).
4. *Requential overhead.* At batch 128 one MNIST epoch has 469 steps; the constant of (4) alone adds $469\kappa \approx 2.4$ kbits, the order of the whole MNIST prequential code (our derivation).
5. *Units and token bookkeeping.* The repositories log nats; Fig. 2c of the founding paper is labelled “MB” and the hard-induction notebook plots nats. The released prequential area integrates over all tokens while the loss covers the label half; the requential sum covers label tokens only. Part of the gap in Fig. 2c may be this factor of two (our derivation from the code).
6. *No seeds.* No epiplexity value in Sections 4 and 5 has an error bar. Related MDL studies use four to eight seeds (Voita and Titov, 2020; Perez et al., 2021; Whitney et al., 2020).
7. *One compute value.* Fig. 3 of the founding paper ranks rule 15 above rule 54 at the lowest compute. Plateau-then-drop tasks such as sparse parity make the area depend on seed and truncation (Barak et al., 2022).

The literature fixes five requirements for a small-scale estimate. The floor of (2) is a held-out or first-pass loss. The standard error of the floor is small against S/M . Code lengths are in bits

over label units, with $\mathcal{D}$ and T stated. Requential values at small batch include the per-message terms of (4). Rankings are checked at more than one compute value.

Testbeds. Table 11 ranks candidate datasets by a variable that sets the amount of structure, a known entropy floor and known learnability. Two designs isolate single effects (our derivation). Symmetric label noise at rate ρ over C classes sets $H(Y | X) = h(\rho) + \rho \log_2(C - 1)$ per example, with h the binary entropy; the raw prequential code grows with this term, while (3) and the requential code should not. A fixed pixel permutation is an exact symmetry of an MLP with i.i.d. initialisation but destroys locality for a convolutional network. S_T should be unchanged for the first observer and change for the second.

Table 11: Candidate testbeds, ranked by our assessment of fitness for MLP estimation on one GPU.

Testbed	Known floor	Structure variable	Main risk	Source
1 ECA, $Y = F^t(X)$	zero	rule; steps t	an MLP lacks locality; add a 1-D convolutional network	Finzi et al. (2026)
2 Markov sources; hidden-row induction	entropy rate	hidden rows	induction needs a sequence model	Edelman et al. (2024)
3 true, noisy, random labels; permuted pixels (MNIST, CIFAR-10, MNIST-1D)	$h(\rho) + \rho \log_2(C - 1)^*$	noise rate; permutation	kilobit signal	Zhang et al. (2017); Greydanus (2020)
4 random hierarchy model	sample complexity known	branching; depth	custom generator	Cagnetta et al. (2023)
5 sparse parity (n, k)	zero	k	plateau then drop	Barak et al. (2022)
6 modular arithmetic	zero	modulus	finite table forces many epochs	Nanda et al. (2023); Gromov (2023)
7 Game of Life	zero	steps	needs strong overparameterisation	Springer and Kenyon (2020)
8 linear congruential against cryptographic generators	$S_T \approx 0$ for a PRG	generator	transformers needed A100-hours	Tao et al. (2025)

Code. The PyTorch trainer of Qiu and Finzi (2026) runs on a CPU as a library call with logging disabled and returns the prequential area per checkpoint (our measurement). An MLP port needs a model class with the same interface, per-class sampling in place of autoregressive generation and areas over label units. A full reproduction of Fig. 3 would take two to four T4-days, a large share of it in data generation on the CPU; a reduced grid (widths 16–128, depths 1–2, data generated on the GPU) takes under two hours per rule (our estimate). Table 12 lists the reusable pieces.

Protocol implied by the literature (our design).

1. Fix $\mathcal{D}$ and three to five values of T for all datasets; report bits, bits per label and S_T/MDL_T .
2. Use one pass over fresh data where the generator allows; otherwise use the first-epoch online loss with a held-out floor, as in (5).
3. Sweep four to six widths with learning rates scaled by 1/fan-in, a constant rate and an EMA; run requential coding with the exact class divergence and report (4) beside $\sum_i \text{KL}_i$.
4. Use at least five seeds with a paired bootstrap; include random labels, constant labels and shuffled targets at matched size.

Questions an MLP project on one GPU can answer.

- *Does an MLP estimate of S_T reproduce the ECA ordering?* Reason: no published estimate uses an MLP or a convolutional network.
- *Does S_T depend on order at small scale?* Reason: only chess shows it.
- *How large is the seed spread of S_T and of the hull?* Reason: no source reports it.
- *Does sequential coding work for classifiers?* Reason: no source validates it; at small batch the per-message terms of (4) are the order of the signal.
- *Which teacher recipe is better, a divergence threshold or iso-loss projection?* Reason: the two papers never compare them.
- *Do the fast estimators agree with S_T ?* Reason: EDL, the reservoir score and the scaling-law proxy have never been applied to the data of a compute sweep.
- *Does S_T predict transfer for small models?* Reason: the evidence is five Pile domains, one counterexample and one null result.

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A Repositories and discrepancies

Table 12: Repositories. Dates are the last commit, or creation where only that is known.

Repository	Framework	Licence	Date	Reusable for small scale
Qiu and Finzi (2026)	PyTorch; JAX for natural data	MIT	8 Mar 2026	<code>soph/train.py</code> : prequential area, requential loop with EMA teacher and threshold; hull code in <code>notebooks/eca_3rules.ipynb</code>
Qiu (2026)	JAX, Flax, Hydra	MIT	14 Jul 2026	<code>train.py</code> with iso-loss projection; noise and repeat controls need no data
Su (2026b)	JAX, Equinox, jaxopt	MIT	25 Aug 2026	<code>scaling_law.py</code> : joint fit and gradient (7); per-domain area in <code>ado.py</code> ; the fit bounds $\log \epsilon \geq 0.5$, too high for low-loss classifiers*
Su (2026a)	PyTorch, Hugging Face	MIT	13 Aug 2026	look-ahead reward <code>_loss_drop</code>
Zhang (2026)	PyTorch	Apache 2.0, non-commercial	not recorded	<code>core.py</code> : score (6); <code>online.py</code> : recursive least squares
Liu (2026)	PyTorch, PEFT	not stated	7 Jul 2026	<code>epiplexity_md1.py</code> : epoch-selection protocol
bossman-lab (2026)	numpy	not stated	24 May 2026	next-token MLP transfer experiment
edirent (2026)	PyTorch	not stated	16 Jun 2026	synthetic controls: silence, tones, noise
lcrh (2026)	TensorFlow.js	not stated	not recorded	reservoir and convolutional observers in the browser
lmxxf (2026)	PyTorch	not stated	not recorded	learnability boundary of pseudorandom generators
RichardScottOZ (2026); Taiyou (2026); tuned-org-uk (2026)	Python	not stated	from Jan 2026	estimator wrappers without published validation

Discrepancies between papers and code, from a comparison of both:

- `eca_3rules.py` uses batch 384 and tag `arxiv_3rules`; App. C.1 states batch 1536 and the notebook filters tag `test` (Qiu and Finzi, 2026).
- `eca_rules.py` disables requential coding and runs 128 automaton steps; Fig. 2c plots requential values and App. C.7 states 48 steps.
- `induction_easy.py` sets a divergence threshold of 0.005; the paper states none.
- The caption of Fig. 8a states 10^9 tokens and the text 5×10^9 . The totals imply about 1.4 bits per character only for 5×10^9 (our derivation).
- Theorem 18 in the appendix keeps the v1 wording “CSPRNG” and “ $-O(1)$ ”.
- The requential paper gives the teacher EMA window as $\max(50, at)$ next to $a = 0.01$; the repository sets $a = 0.1$ for the teacher (Qiu, 2026).
- App. C.1 of Su et al. (2026) names Pile domains in a Common Pile analysis. The EpiGen mixture configurations replace 25, 50 and 100 per cent of learner data by real text, while the paper describes 25 and 50 per cent synthetic (Su, 2026a).
- App. D of Zhang and Levin (2026) places rule 41 second, Fig. 2 rule 25. The README lists a table column absent from the paper.

- The arXiv EDL preprint credits one diagnostic example with “exactly” $\log |\mathcal{Y}|$ bits for label alphabet $\mathcal{Y}$ in Sec. 5.2 and “up to” $\log m$ bits for m hypotheses in App. B.2 (Donoway et al., 2026).